

1 Classical mechanics

1.1 Lagrangian formulation

[1.] The Lagrangian of the system consists of three terms: the kinetic energy T_{kin} , the potential energy of the vertical springs $V_{\parallel}$ and the potential energy of the horizontal springs $V_{\perp}$:

$$L = T_{\text{kin}} - V_{\parallel} - V_{\perp}$$

The kinetic term is given by

$$T_{\text{kin}} = \sum_{j,k=1}^N \frac{1}{2} \mu \dot{q}_{jk}^2,$$

the vertical springs' potential energy can be written as

$$V_{\parallel} = \sum_{j,k=1}^N \frac{1}{2} \lambda q_{jk}^2.$$

The potential energy of the horizontal springs can be calculated by first summing over all springs in the x direction and then in the y direction:

$$V_{\perp} = \sum_{k=1}^N \sum_{j=1}^N \frac{1}{2} \kappa (q_{j+1k} - q_{jk})^2 + \sum_{j=1}^N \sum_{k=1}^N \frac{1}{2} \kappa (q_{jk+1} - q_{jk})^2$$

Overall, we obtain the following formula for the Lagrangian:

$$L = \sum_{j,k=1}^N \frac{1}{2} [\mu \dot{q}_{jk}^2 - \lambda q_{jk}^2 - \kappa (q_{j+1k} - q_{jk})^2 - \kappa (q_{jk+1} - q_{jk})^2]$$

[2.] From the Lagrange equations of the second kind, we derive the following equations of motion for each q_{jk} :

$$\mu \ddot{q}_{jk} + \lambda q_{jk} + \kappa [4q_{jk} - q_{j+1k} - q_{j-1k} - q_{jk+1} - q_{jk-1}] = 0$$

[3.] Using the symmetry of the model, specifically translational invariance with

periodic boundary conditions, we search for the solution to the equations of motion in the form of a discrete Fourier transform

$$q_{jk}(t) = \sum_{m,n=1}^N \left(c_{mn} \exp \left[\frac{2\pi i}{N} (mj + nk) - i\omega_{mn} t \right] + c_{mn}^* \exp \left[-\frac{2\pi i}{N} (mj + nk) + i\omega_{mn} t \right] \right),$$

where the complex conjugate part of the solution ensures the reality of q_{jk} .

[4.] By plugging this solution into the equations of motion, we obtain the following equation

$$\sum_{m,n=1}^N c_{mn} \exp \left[\frac{2\pi i}{N} (mj + nk) - i\omega_{mn} t \right] \left[-\mu\omega_{mn}^2 + \lambda + \kappa \left[4 - \exp \left(\frac{2\pi i m}{N} \right) - \exp \left(-\frac{2\pi i m}{N} \right) - \exp \left(\frac{2\pi i n}{N} \right) - \exp \left(-\frac{2\pi i n}{N} \right) \right] \right] + \text{c.c.} = 0.$$

For the left-hand side to be equal to zero for all j , k , and t , the last term (the last bracket) of the left-hand side must equal zero, giving us, after a few algebraic manipu-

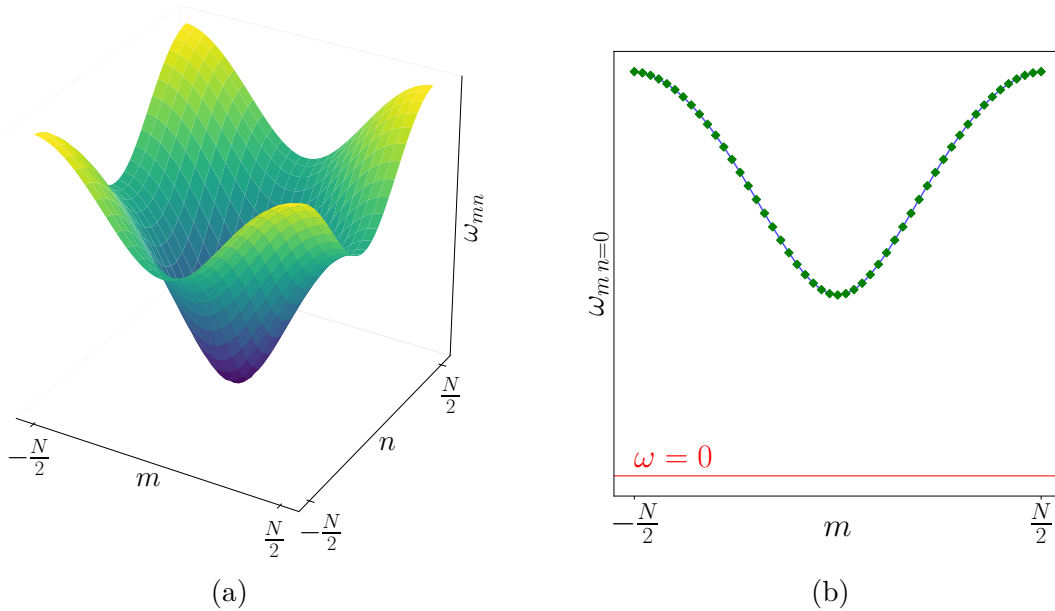


Figure 1: This graph shows the dispersion relation ω_{mn} . The first graph (left) shows the full dependence on both m and n , while the second graph (right) shows a slice for $n = 0$. The graphs were plotted for $N = 50$, $\mu = 1$, $\lambda = 1$, and $\kappa = 1$.

lations, the dispersion relation:

$$\omega_{mn}^2 = \frac{\lambda}{\mu} + \frac{4\kappa}{\mu} \sin^2\left(\frac{\pi m}{N}\right) + \frac{4\kappa}{\mu} \sin^2\left(\frac{\pi n}{N}\right)$$

Given that the $\sin^2$ functions are always non-negative, and assuming that $\lambda, \kappa, \mu \geq 0$, the only case in which ω_{mn} , and therefore the energy, can be zero is when $\lambda = 0$. In addition, $q_{jk}(t) \equiv 0 \forall j, k$ trivially solves the classical equations of motion with zero energy.

1.2 Hamiltonian formulation

[1.] Firstly, we define the canonical momenta as

$$p_{jk} := \frac{\partial L}{\partial \dot{q}_{jk}} \equiv \mu \dot{q}_{jk}.$$

The Hamiltonian is then given by the Legendre transform of the Lagrangian. We obtain the expression:

$$\begin{aligned} H &= \sum_{m,n=1}^N \frac{\partial L}{\partial \dot{q}_{mn}} \dot{q}_{mn} - L \bigg|_{\dot{q}_{jk} = \frac{1}{\mu} p_{jk}} \\ &= \sum_{j,k=1}^N \frac{1}{2} \left[\frac{1}{\mu} p_{jk}^2 + \lambda q_{jk}^2 + \kappa (q_{j+1,k} - q_{jk})^2 + \kappa (q_{j,k+1} - q_{jk})^2 \right] \end{aligned}$$

Finally, we define the canonical Poisson brackets as

$$\{f, g\} := \sum_{j,k=1}^N \left[\frac{\partial f}{\partial q_{jk}} \frac{\partial g}{\partial p_{jk}} - \frac{\partial f}{\partial p_{jk}} \frac{\partial g}{\partial q_{jk}} \right]$$

It is then straightforward to prove that they satisfy the following relations:

$$\{q_{jk}, q_{mn}\} = \{p_{jk}, p_{mn}\} = 0, \quad \{q_{jk}, p_{mn}\} = \delta_{jm} \delta_{kn}$$

[2.] Similarly to the previous exercise, the discrete Fourier transform, respecting the periodic boundary condition, will be the key to finding the desired set of modes

(a, a^*) which diagonalizes the Hamiltonian. Firstly, we define the canonical conjugate momenta and coordinates in the Fourier space:

$$\begin{aligned}\tilde{q}_{jk} &= \sum_{m,n=1}^N q_{mn} \exp \left[-\frac{2\pi i}{N}(jm + kn) \right] \\ \tilde{p}_{jk} &= \sum_{m,n=1}^N p_{mn} \exp \left[-\frac{2\pi i}{N}(jm + kn) \right]\end{aligned}$$

Using the dispersion relation from the first exercise, we can rewrite the Hamiltonian in the following form

$$H = \frac{1}{N^2} \sum_{j,k=1}^N \left[\frac{1}{2\mu} \tilde{p}_{jk}^2 + \frac{1}{2} \mu \omega_{jk}^2 \tilde{q}_{jk}^2 \right].$$

Finally, we define the quasiparticle modes a and a^* as

$$\begin{aligned}a_{jk} &= \frac{1}{\sqrt{2\mu\omega_{jk}}} [\mu\omega_{jk}\tilde{q}_{jk} + i\tilde{p}_{jk}] = \frac{1}{\sqrt{2\mu\omega_{jk}}} \sum_{m,n=1}^N \exp \left[-\frac{2\pi i}{N}(jm + kn) \right] [\mu\omega_{jk}q_{mn} + ip_{mn}] \\ a_{jk}^* &= \frac{1}{\sqrt{2\mu\omega_{jk}}} [\mu\omega_{jk}\tilde{q}_{jk}^* - i\tilde{p}_{jk}^*] = \frac{1}{\sqrt{2\mu\omega_{jk}}} \sum_{m,n=1}^N \exp \left[+\frac{2\pi i}{N}(jm + kn) \right] [\mu\omega_{jk}q_{mn} - ip_{mn}],\end{aligned}$$

which allows us to write the Hamiltonian in the following form:

$$H = \frac{1}{N^2} \sum_{j,k=1}^N \omega_{jk} a_{jk}^* a_{jk}$$

[3.] After some mostly algebraic work (which I omit for clarity), we can derive the following Poisson brackets for the quasiparticle modes:

$$\{a_{jk}, a_{mn}\} = \{a_{jk}^*, a_{mn}^*\} = 0, \quad \{a_{jk}, a_{mn}^*\} = -iN^2 \delta_{jm} \delta_{kn}$$

Using these Poisson brackets, we can derive the following Poisson bracket:

$$\begin{aligned}\{a_{jk}, H\} &= \frac{1}{N^2} \sum_{m,n=1}^N \omega_{jk} \{a_{jk}, a_{mn}^* a_{mn}\} = \frac{1}{N^2} \sum_{m,n=1}^N \omega_{jk} [\{a_{jk}, a_{mn}^*\} a_{mn} + a_{mn}^* \{a_{jk}, a_{mn}\}] \\ &= \frac{1}{N^2} \sum_{m,n=1}^N \omega_{jk} [-iN^2 \delta_{jm} \delta_{kn} a_{mn} + a_{mn}^* \cdot 0] = -i\omega_{jk} a_{jk}\end{aligned}$$

By taking the complex conjugate of the previous equation, we automatically obtain $\{a_{jk}^*, H\} = i\omega_{jk}a_{jk}^*$.

Using the canonical equations of motion, we obtain

$$\dot{a}_{jk} = \{a_{jk}, H\} = -i\omega_{jk}a_{jk}, \quad \dot{a}_{jk}^* = \{a_{jk}^*, H\} = i\omega_{jk}a_{jk}^*.$$

These equations can be easily solved:

$$a_{jk}(t) = \alpha_{jk} \exp[-i\omega_{jk}t], \quad a_{jk}^*(t) = \alpha_{jk}^* \exp[i\omega_{jk}t]$$

When we compare the solution obtained using the Hamiltonian formalism with the one derived from the Lagrangian formalism, by applying the inverse Fourier transform and solving for q_{jk} , we see that the coefficients α_{jk} and c_{jk} differ only in normalization:

$$\alpha_{jk} = N^2 \sqrt{2\mu\omega_{jk}} c_{jk}$$

2 Continuum limit of the lattice model

[1.] Firstly, we derive the formulae that allow us to transition from differences and sums to derivatives and integrals. For the derivatives, it is straightforward to see that

$$q_{j+1,k} - q_{j,k} \equiv \varphi_{[(j+1)r,kr]} - \varphi_{[jr,kr]} = \frac{\varphi_{[(j+1)r,kr]} - \varphi_{[jr,kr]}}{r} r \approx r \partial_x \varphi([jr, kr]).$$

For the sums, we introduce the area element r^2 , which allows us to transition to integrals in the continuous limit:

$$\sum_{j,k=1}^N f_{jk} r^2 \rightarrow \int d\mathbf{x} f(\mathbf{x}) \quad \Rightarrow \quad \sum_{j,k=1}^N f_{jk} \rightarrow \frac{1}{r^2} \int d\mathbf{x} f(\mathbf{x})$$

Overall, using the obtained formulae, we can rewrite the Lagrangian in terms of the smooth field φ :

$$L = \int d\mathbf{x} \frac{1}{2} \left[\frac{\mu}{r^2} \dot{\varphi}^2 - \frac{\lambda}{r^2} \varphi^2 - \kappa (\partial \varphi)^2 \right],$$

where we have simplified the notation $\varphi \equiv \varphi(\mathbf{x})$ and defined $\partial := [\partial_x, \partial_y]$.

We see that this expression diverges as $r \rightarrow 0$. Therefore, we have to renormalize the parameters μ , λ , and κ as follows:

$$\mu \rightarrow r^2 \tilde{\mu}, \quad \lambda \rightarrow r^2 \tilde{\lambda}, \quad \kappa \rightarrow \tilde{\kappa}.$$

From this, we see that the κ parameter is already properly normalized. After doing this, we obtain the following expression for the Lagrangian functional:

$$L[\varphi, \dot{\varphi}](t) = \int d\mathbf{x} \frac{1}{2} \left[\tilde{\mu} \dot{\varphi}^2 - \tilde{\lambda} \varphi^2 - \tilde{\kappa} (\partial \varphi)^2 \right].$$

[2.] To canonically normalize the field, we choose:

$$\varphi \rightarrow \frac{1}{\sqrt{\tilde{\kappa}}} \phi$$

and obtain the Lagrangian in the following form:

$$L[\phi, \dot{\phi}](t) = \int d\mathbf{x} \left[\frac{\tilde{\mu}}{2\tilde{\kappa}} \dot{\phi}^2 - \frac{\tilde{\lambda}}{2\tilde{\kappa}} \phi^2 - \frac{1}{2} (\partial \phi)^2 \right].$$

Now we define the action functional:

$$S[\phi] := \int dt L[\phi, \dot{\phi}](t) \equiv \int dx^\mu \mathcal{L}(\phi, \partial_\mu \phi),$$

where $\mathcal{L}$ is the Lagrangian density. By varying the action:

$$\delta S[\phi] = \int dx^\mu \delta \mathcal{L} = \int dx^\mu \left[\frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta (\partial_\mu \phi) \right] = \int dx^\mu \delta \phi \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right],$$

where we have used the periodic boundary conditions to eliminate the boundary terms, and by setting $\delta S[\phi] \stackrel{!}{=} 0$, we obtain the Euler-Lagrange equation

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} = 0.$$

Using the previously obtained Lagrangian density

$$\mathcal{L} = \frac{\tilde{\mu}}{2\tilde{\kappa}} \dot{\phi}^2 - \frac{\tilde{\lambda}}{2\tilde{\kappa}} \phi^2 - \frac{1}{2} (\partial \phi)^2,$$

and the following relations

$$\frac{\partial \mathcal{L}}{\partial \phi} = -\frac{\tilde{\lambda}}{\tilde{\kappa}}\phi, \quad \frac{\partial \mathcal{L}}{\partial(\partial_j \phi)} = -\partial_j \phi, \quad \frac{\partial \mathcal{L}}{\partial(\partial_t \phi)} = \frac{\tilde{\mu}}{\tilde{\kappa}}\partial_t \phi,$$

we derive the equations of motion for the field ϕ

$$\frac{\tilde{\mu}}{\tilde{\kappa}}\partial_t^2 \phi + \partial^j \partial_j \phi + \frac{\tilde{\lambda}}{\tilde{\kappa}}\phi = 0,$$

where $\partial^j \partial_j \equiv -\Delta \equiv -\sum_j \partial_j^2$.

[3.] From the equations of motion for the field ϕ , we can clearly identify the speed of sound c and the mass m :

$$\frac{1}{c^2} = \frac{\tilde{\mu}}{\tilde{\kappa}} \quad \text{and} \quad m^2 = \frac{\tilde{\lambda}}{\tilde{\kappa}}.$$

[4.] By setting the speed of sound $c = 1$, we can rewrite the equations of motion into a simpler form:

$$(\partial^\mu \partial_\mu + m^2) \phi = 0,$$

which resembles the Klein-Gordon equation.

The solution of the Klein-Gordon equation can be expressed in terms of plane waves. A general solution can thus be written in the following form using the Fourier transform (which corresponds to the discrete Fourier series in the lattice case):

$$\phi(\mathbf{x}, t) = \frac{1}{(2\pi)^2} \int d\mathbf{p} [c(\mathbf{p}) \exp[i(\mathbf{p} \cdot \mathbf{x} - E(\mathbf{p})t)] + c^*(\mathbf{p}) \exp[-i(\mathbf{p} \cdot \mathbf{x} - E(\mathbf{p})t)],$$

where $E(\mathbf{p})$ is defined using the relativistic dispersion relation:

$$E^2(\mathbf{p}) = p^2 + m^2.$$

By identifying the terms in the exponential of the solution for the field with the solution for the discrete lattice, we get:

$$\omega_{jk} \leftrightarrow E(\mathbf{p}),$$

and using $x_i \leftrightarrow jr$, we obtain:

$$p_i \leftrightarrow \frac{2\pi m}{Nr} = \frac{2\pi m}{L}.$$

Also, $c_{jk} \leftrightarrow c(\mathbf{p})$.

Using these relations, we can substitute them into the dispersion relation for the discrete case:

$$\begin{aligned}\omega_{mn}^2 &= \frac{\lambda}{\mu} + \frac{4\kappa}{\mu} \sin^2\left(\frac{\pi m}{N}\right) + \frac{4\kappa}{\mu} \sin^2\left(\frac{\pi n}{N}\right) \\ &\quad \updownarrow \\ E^2(\mathbf{p}) &= \lim_{r \rightarrow 0} \left[\frac{\tilde{\lambda}}{\tilde{\kappa}} \frac{\tilde{\kappa}}{\tilde{\mu}} + \frac{4\tilde{\kappa}}{\tilde{\mu}} \frac{1}{r^2} \sin^2\left(\frac{p_x r}{2}\right) + \frac{4\tilde{\kappa}}{\tilde{\mu}} \frac{1}{r^2} \sin^2\left(\frac{p_y r}{2}\right) \right] \\ &= m^2 + p_x^2 + p_y^2 = m^2 + p^2,\end{aligned}$$

where we have used the fact that the speed of sound is equal to one. Therefore, we have shown that both dispersion relations are, in a sense, equivalent.